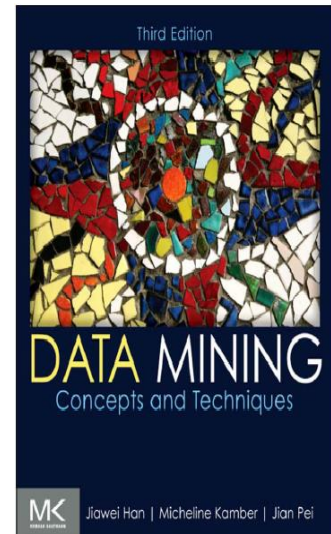


**Data Mining & Business Intelligence (IT647)**  
**Selected Topics In Data Science (IT688)**  
**Information Technology Department**  
**College Of Computer**  
**Qassim University**  
**DR.MONA ALSALEEM**

## Data Mining

### — Chapter 1 — Introduction

Text Book: “Data Mining: Concepts and Techniques”, Third Edition



Why Data Mining?

History Of Data Mining

What Is Data Mining?

What Kinds of Data Can Be Mined?

What Kinds of Patterns Can Be Mined?

What Kinds of Technologies Are Used?

What Kinds of Applications Are Targeted?

Summary

# Why Data Mining?

- The Explosive Growth of Data: from terabytes to petabytes
- Data collection and data availability
- Automated data collection tools, database systems, Web, computerized society

**We are drowning in data, but starving for knowledge!**

“Necessity is the mother of invention”—***Data mining***—Automated analysis of massive data sets

# History Of Data Mining

- 1960s: Data collection, database creation, IMS and network DBMS
- 1970s: Relational data model, relational DBMS implementation
- 1980s: RDBMS, advanced data models (extended-relational, OO, deductive, etc.)
- 1990s-now: Data mining, data warehousing, multimedia databases, and Web databases

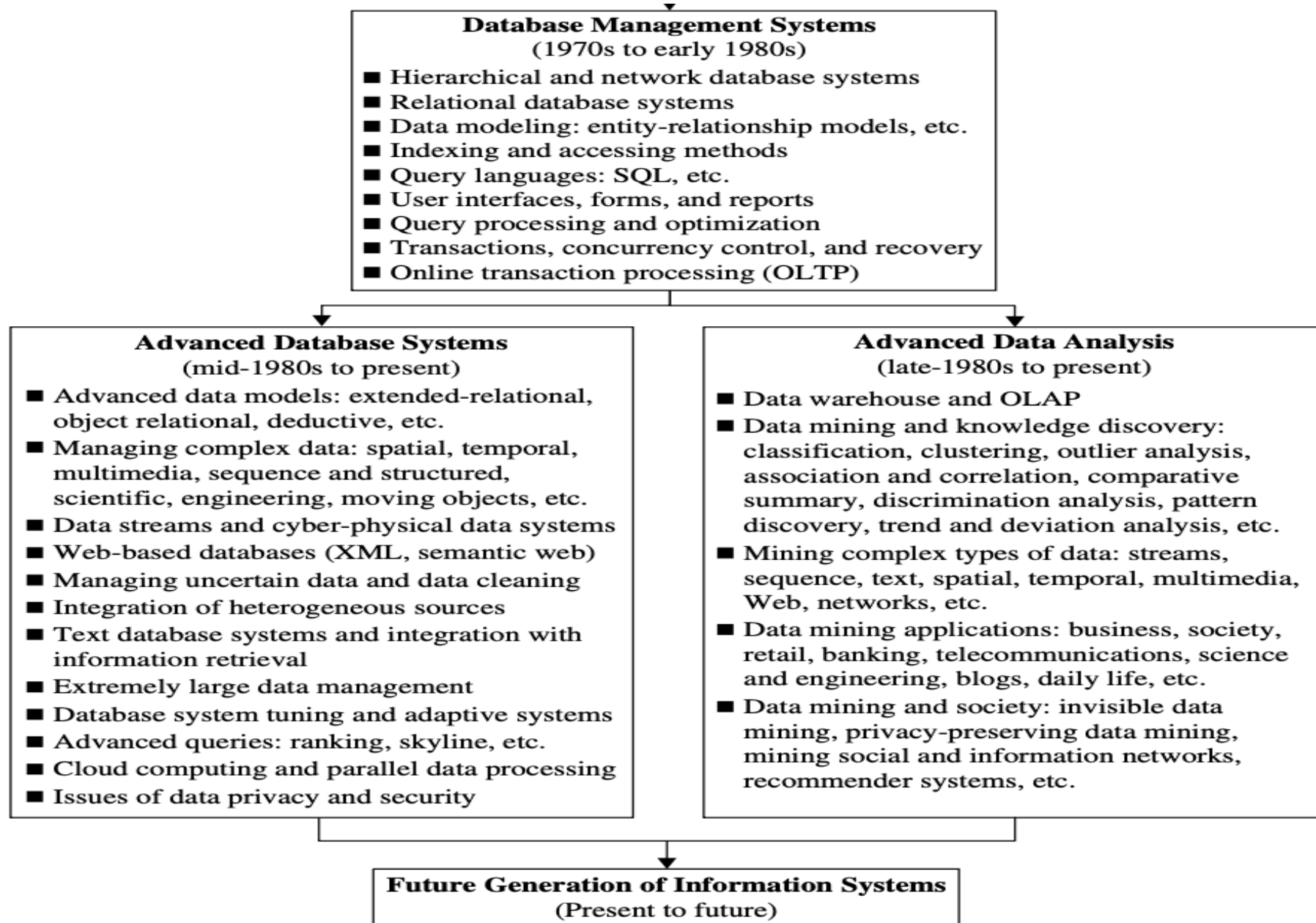
The ability to store and manage petabytes of data online :

***Data mining* is a major new challenge!**

- 2000s

Stream data management and Data mining with its applications

Web technology (XML, data integration) and global information systems



# What Is Data Mining?

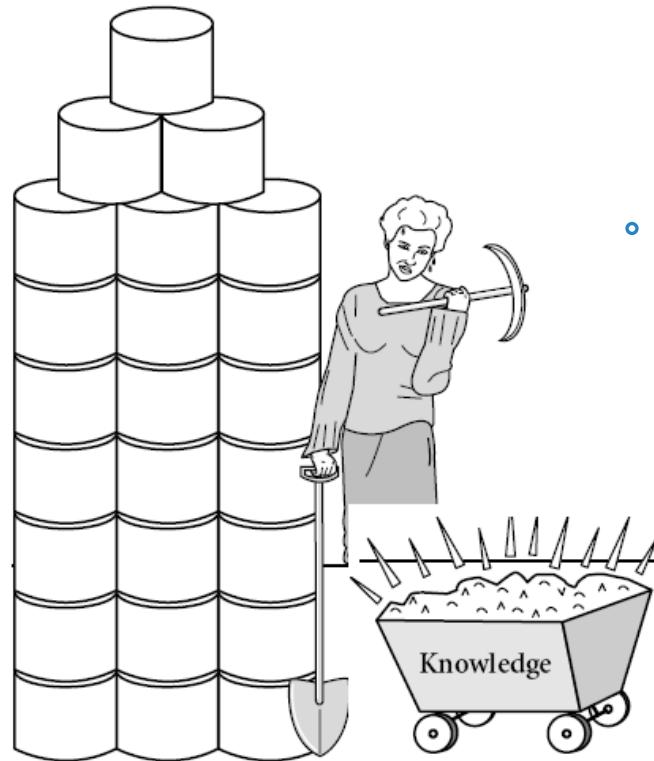
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- Data mining (knowledge discovery from data)
  - Simply stated, data mining refers to extracting or “mining” knowledge from large amounts of data”.
- Data mining have been more appropriately named *“Knowledge mining from data”*.



Data mining—searching for knowledge (interesting patterns) in your data.

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How can I analyze my data?

We have a lot of data but lack the information and knowledge

Data Mining is a part of the knowledge Discovery Process, Knowledge Discovery from Data abbreviated as **(KDD)**



# Knowledge Discovery (KDD) Process

Typically, the steps of a KDD process are:

- Data cleaning (to remove noise and inconsistent data)
- Data integration (where multiple data sources may be combined)
- Data selection (where data relevant to the analysis task are retrieved from the database)
- Data transformation (converting data into a form more appropriate for mining by performing summary or aggregation operations)
- **Data mining** (an essential process where intelligent methods are applied to extract data patterns)
- Pattern evaluation (to identify the interesting patterns representing knowledge)
- Knowledge presentation (where visualization and knowledge presentation techniques are used to present mined knowledge to users)

# Knowledge Discovery (KDD) Process



An Overview of the Steps That Compose the KDD Process

# Data Mining Scope

- **Finance and business:**  
Fraud detection, Market forecasting  
Basket analysis, Product targeting, Efficient mailing
- **Engineering:**  
Process modeling and optimization  
Machine diagnostics, Predictive maintenance
- **Internet:**  
Text mining, Intelligent query answering  
Web access analysis, Site personalization
- **Medical Informatics and others**

# Data Mining: On What Kinds of Data?

## Database-oriented data sets and applications

- Relational database, data warehouse, transactional database
- Object-relational databases, Heterogeneous databases and legacy databases.

## Advanced data sets and advanced applications

- Data streams
- Structure data, graphs, social networks and information networks
- Spatial data and spatiotemporal data
- Multimedia database
- The World-Wide Web

# Data Mining: On What Kinds of Data?

## Database Data

- database system, also called a database management system (DBMS), consists of a collection of interrelated data, known as a database, and a set of software programs to manage and access the data.
- A relational database is a collection of tables, each of which is assigned a unique name.
- Each table consists of a set of attributes (columns or fields) and usually stores a large set of tuples (records or rows).
- Each tuple in a relational table represents an object identified by a unique key and described by a set of attribute values.

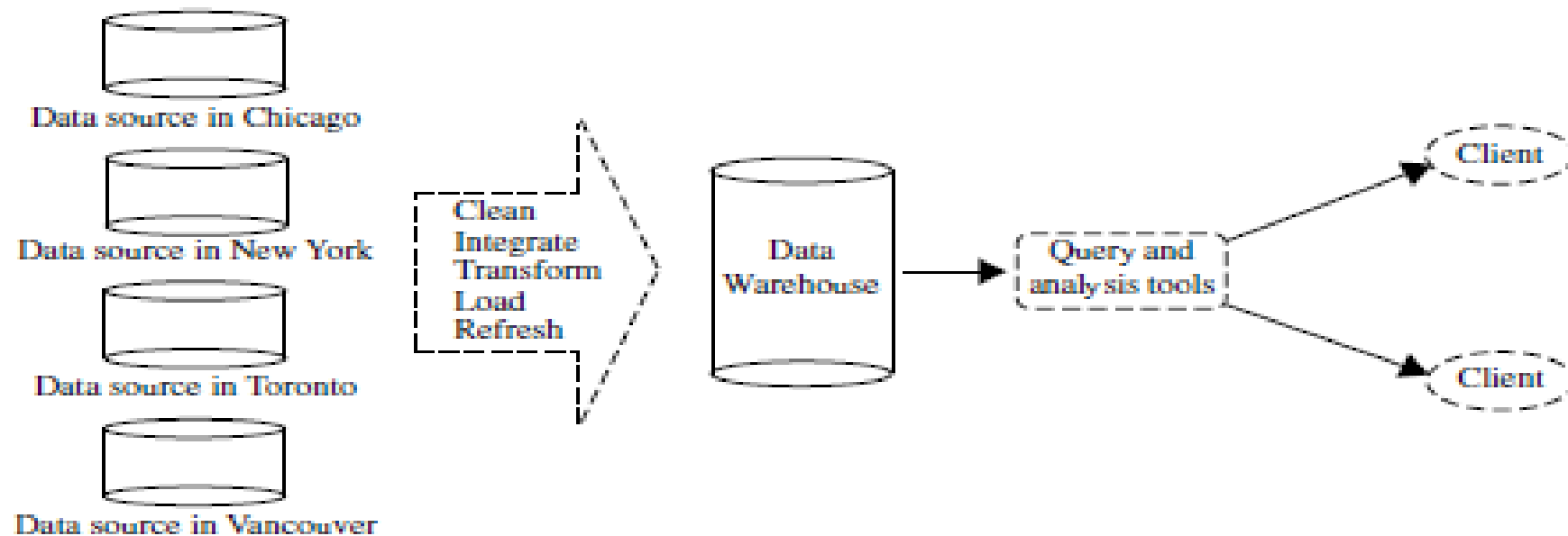
# Data Mining: On What Kinds of Data?

## Data Warehouses

- A data warehouse is a repository of information collected from multiple sources, stored under a unified schema, and usually residing at a single site.
- Data warehouses are constructed via a process of data cleaning, data integration, data transformation, data loading, and periodic data refreshing.
- It is usually modeled by a multidimensional data structure, called a data cube, in which each dimension corresponds to an attribute or a set of attributes in the schema, and each cell stores the value of some aggregate measure.

# Data Mining: On What Kinds of Data?

## Data Warehouses



*Typical framework of a data warehouse for AllElectronics.*

# Data Mining: On What Kinds of Data?

## Transactional Data

- Each record in a transactional database captures a transaction, such as a customer's purchase, a flight booking, or a user's clicks on a web page.
- A transaction typically includes a unique transaction identity number (trans ID) and a list of the items making up the transaction, such as the items purchased in the transaction.

| <i>trans_ID</i> | <i>list_of_item_IDs</i> |
|-----------------|-------------------------|
| T100            | I1, I3, I8, I16         |
| T200            | I2, I8                  |
| ...             | ...                     |

*Fragment of a transactional database for sales at AllElectronics*



# What Kinds of Patterns Can Be Mined?

**There are a number of data mining functionalities. These include:**

- characterization and discrimination
- The mining of frequent patterns, associations, and correlations;
- classification and regression;
- clustering analysis
- outlier analysis

Data mining functionalities are used to specify the kinds of patterns to be found in data mining tasks.

In general, such tasks can be classified into two categories: descriptive and predictive.

# What Kinds of Patterns Can Be Mined?

## Data Characterization

- Data characterization is a summarization of the general characteristics or features of a target class of data.
- For example, to study the characteristics of software products with sales that increased by 10% in the previous year

## Data Discrimination

- Data discrimination is a comparison of the general features of the target class data objects against the general features of objects from one or multiple contrasting classes.
- For example, a user may want to compare the general features of software products with sales that increased by 10% last year against those with sales that decreased by at least 30% during the same period.

# What Kinds of Patterns Can Be Mined?

## Mining Frequent Patterns, Associations, and Correlations

- Frequent patterns, as the name suggests, are patterns that occur frequently in data.
- Mining frequent patterns leads to the discovery of interesting associations and correlations within data.

- **Example of association rule :**

*buys(X, “computer”)  $\Rightarrow$  buys(X, “software”) [support = 1%, confidence = 50%],*

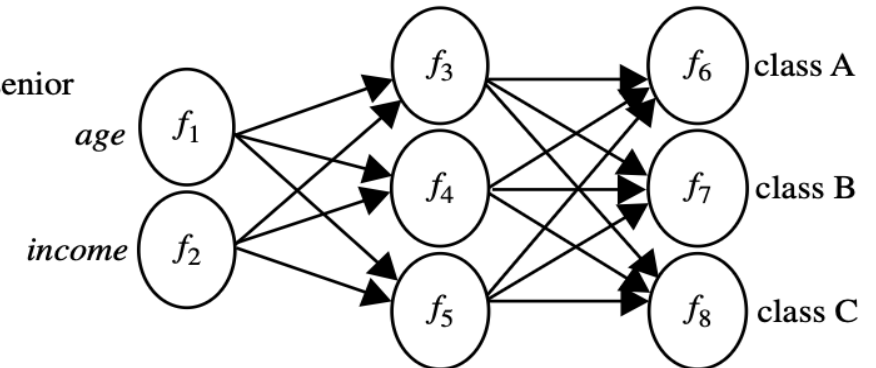
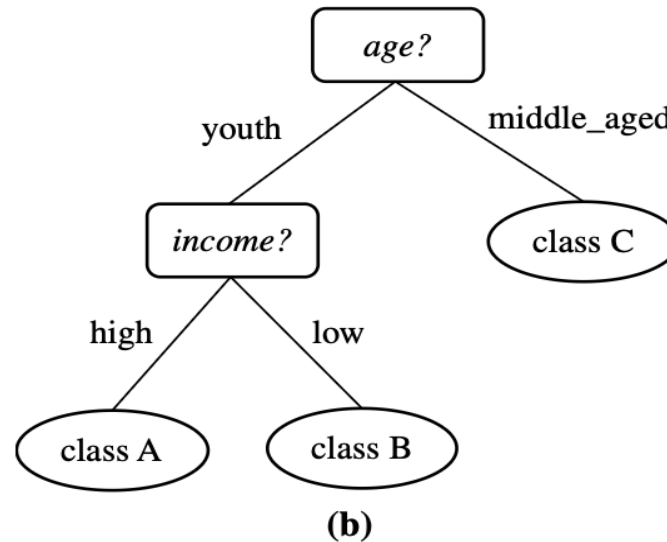
- Typically, association rules are discarded as uninteresting if they do not satisfy both a minimum support threshold and a minimum confidence threshold.
- Additional analysis can be performed to uncover interesting statistical correlations between associated attribute–value pairs.

## Classification

- The process of finding a model (or function) that describes and distinguishes data classes or concepts.
- The model are derived based on the analysis of a set of training data (i.e., data objects for which the class labels are known).
- The model is used to predict the class label of objects for which the the class label is unknown.

$age(X, \text{"youth"}) \text{ AND } income(X, \text{"high"}) \longrightarrow class(X, \text{"A"})$   
 $age(X, \text{"youth"}) \text{ AND } income(X, \text{"low"}) \longrightarrow class(X, \text{"B"})$   
 $age(X, \text{"middle\_aged"}) \longrightarrow class(X, \text{"C"})$   
 $age(X, \text{"senior"}) \longrightarrow class(X, \text{"C"})$

(a)



A classification model can be represented in various forms: (a) IF-THEN rules, (b) a decision tree, or (c) a neural network.

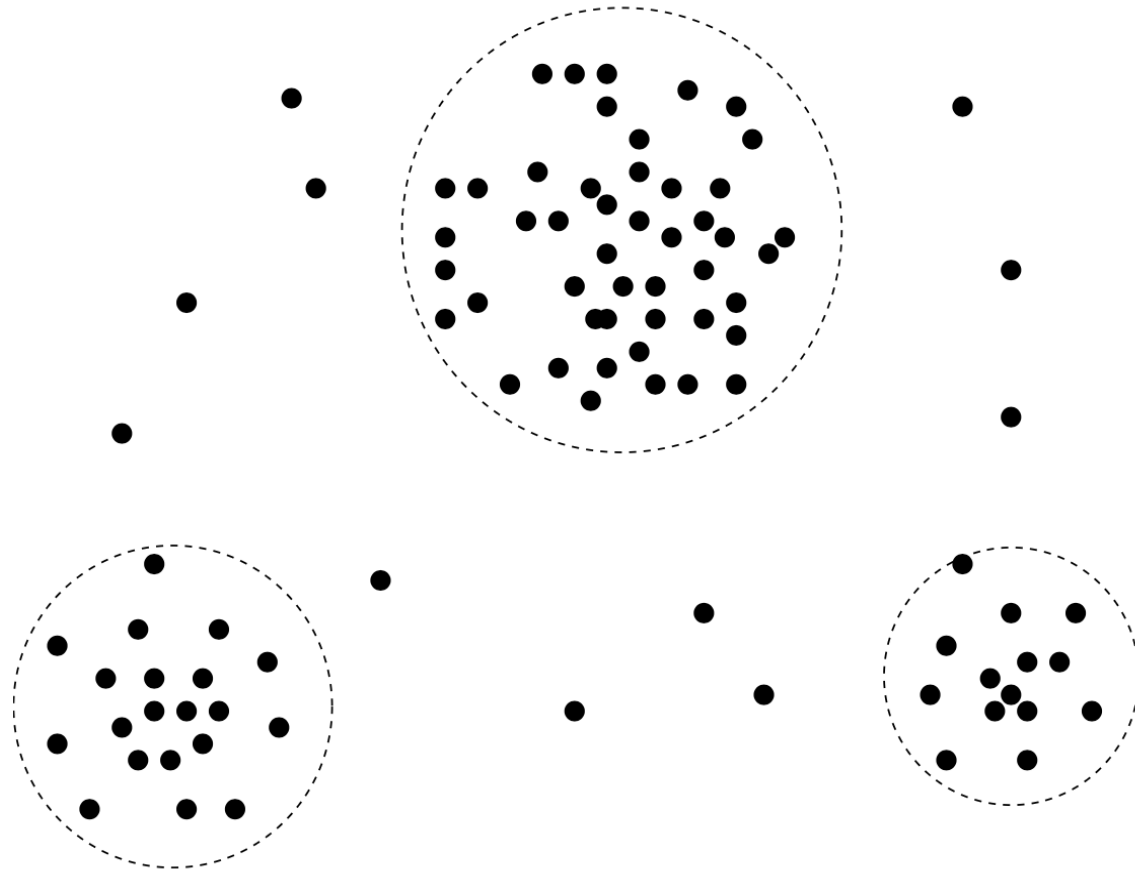
# What Kinds of Patterns Can Be Mined?

## Regression

Regression analysis is a statistical methodology that is most often used for numeric prediction.

## Clustering

clustering analyzes data objects without consulting class labels.

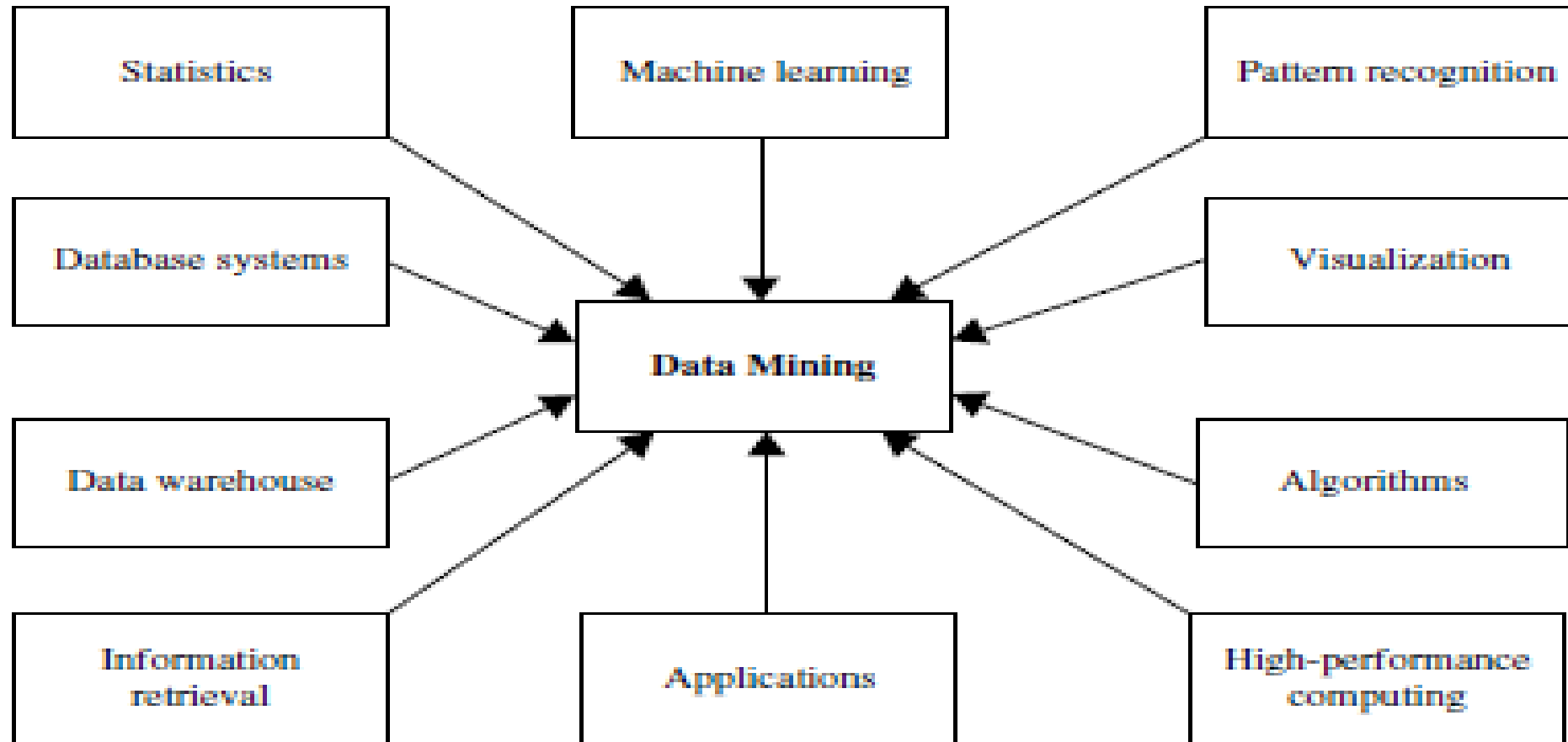


# What Kinds of Patterns Can Be Mined?

## Outlier analysis

- A data set may contain objects that do not comply with the general behavior or model of the data. These data objects are outliers.
- Many data mining methods discard outliers as noise or exceptions. However, in some applications (e.g., fraud detection) events can be more interesting than the more regularly occurring

# Which Technologies Are Used?



*Data mining adopts techniques from many domains.*

# Which Kinds of Applications Are Targeted?

## Where there are data, there are data mining applications

- As a highly application-driven discipline, data mining has seen great successes in many applications;
- It is impossible to enumerate all applications where data mining plays a critical role;
- Presentations of data mining in knowledge-intensive application domains require more in-depth treatment.



# Which Kinds of Applications Are Targeted?

## *Business Intelligence*

- Business intelligence (BI) technologies provide historical, current, and predictive views of business operations.
- Clearly, data mining is the core of business intelligence. Online analytical processing tools in business intelligence rely on data warehousing and multidimensional data mining.
- Classification and prediction techniques are the core of predictive analytics in business intelligence

# Which Kinds of Applications Are Targeted?

## *Web Search Engines*

- A Web search engine is a specialized computer server that searches for information on the Web.
- Various data mining techniques are used in all aspects of search engines, ranging from :
  - ❖ crawling (deciding which pages should be crawled and the crawling frequencies),
  - ❖ indexing (selecting pages to be indexed and deciding to which extent the index should be constructed)
  - ❖ searching (deciding how pages should be ranked,)

# Applications of Data Mining

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- Web page analysis: from web page classification, clustering to PageRank
- Collaborative analysis & recommender systems
- Basket market data analysis to targeted marketing
- Biological and medical data analysis: classification, cluster analysis (microarray data analysis), biological sequence analysis, biological network analysis
- Data mining and software engineering
- From major dedicated data mining systems/tools (e.g., SAS, MS SQL-Server Analysis Manager, Oracle Data Mining Tools) to invisible data mining

# Summary

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- Data mining: Discovering interesting patterns and knowledge from massive amount of data
- A natural evolution of science and information technology, in great demand, with wide applications
- A KDD process includes data cleaning, data integration, data selection, transformation, data mining, pattern evaluation, and knowledge presentation
- Mining can be performed in a variety of data.
- Data mining functionalities: characterization, discrimination, association, classification, clustering, trend and outlier analysis, etc.
- Data mining technologies and applications

# Assignment

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AI Magazine Volume 17 Number 3 (1996) (© AAAI)

## From Data Mining to Knowledge Discovery in Databases

*Usama Fayyad, Gregory Piatetsky-Shapiro, and Padhraic Smyth*

This paper is:

- A foundational reference in data science
- Often cited in several research papers
- explain the methodology behind data-driven discovery

### Your assignment :

- Summarize the paper in 2 pages
- Similar work will have zero

Thank you

